

LG-CISH: Lossless Language-Guided Coverless Image Steganography via CLIP Semantic Hashing and Base- N Positional Coding

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Abstract

Embedding-based image steganography perturbs cover pixels and is therefore detectable by modern statistical and CNN steganalysers. We present **LG-CISH**, a fully *lossless coverless* scheme that hides a secret message in a sequence of *unmodified* natural images: the identity and order of the images carry the data. LG-CISH combines two ideas. First, each codebook image is represented by a normalised CLIP embedding, so recovery is a robust semantic nearest-neighbour lookup that survives JPEG, resizing, noise, and aggressive cropping—unlike perceptual hashing. Second, base- N positional coding treats the encrypted, integrity-protected payload as a single integer written in radix N , giving the information-theoretic $\log_2 N$ bits per image. A thin zlib/AES-256/CRC-32 layer adds compression, confidentiality, and tamper detection, and two optional cover-shaping layers—a distinct-image *permutation* mode and an LLM-guided *alias* layer—reshape the cover without altering the encoded bits. On a 40-image codebook drawn from the UCID, Kodak, and USC-SIPI benchmark suites (5.322 bits/image), LG-CISH achieves bit-exact reconstruction ($\text{BER} = 0$), 100% accuracy across twenty-one channel attacks (JPEG down to quality 20, noise, downscaling to 25%, centre-cropping to 60%, and WebP conversion), chance-level steganalysis detectability (47.5% vs. 97.5% for LSB), and a statistically significant geometric-robustness advantage over perceptual hashing (paired Wilcoxon signed-rank, $p = 7.8 \times 10^{-3}$). We further quantify cover plausibility with an LLM judge, and analyse the capacity/undetectability and theme/plausibility trade-offs and the database-bound capacity limit of the coverless paradigm.

Keywords: coverless steganography; information hiding; CLIP; semantic hashing; base- N coding; robustness; steganalysis resistance.

1 Introduction

Image steganography conceals a secret message inside a cover image so that its very existence is hidden from an observer. Classical embedding methods — least significant bit (LSB) substitution [7, 8], discrete cosine transform (DCT) and discrete wavelet transform (DWT) coefficient modification [10], and learned auto-encoders [11, 12] — all perturb the cover pixels. Those perturbations, however small, leave a statistical fingerprint that modern convolutional steganalysers such as XuNet [5] and SRNet [4] can detect with high accuracy.

Coverless image steganography removes the attack surface entirely: no pixel is ever modified [2, 13]. Instead, the secret is conveyed by *selecting* or *generating* ordinary images whose intrinsic features map to the message bits. Because the transmitted images are statistically indistinguishable from natural images, pixel-domain steganalysis is reduced to random guessing. The central difficulty of the selection (mapping) paradigm is *reliable recovery*: the receiver must map each received image back to the exact index the sender intended, even after the image has been re-compressed, resized, or noised by a real transport channel (e.g. a messaging platform). Robust hashing [2, 16] and hand-crafted features such as SURF [17] have been used for this mapping, but they remain sensitive to geometric distortion.

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We propose **LG-CISH** (Language-Guided Coverless Image Steganography via Hashing), a fully *lossless* coverless scheme built on two ideas:

1. **Semantic hashing with a vision–language model.** Each codebook image is represented by a normalised CLIP [1] embedding. Recovery is a nearest-neighbour lookup in this semantic space. Unlike a perceptual hash, the CLIP embedding encodes *visual meaning*, so it survives JPEG, resizing, noise, and aggressive cropping (Section 4).
2. **Base- N positional coding.** With a codebook of N images we treat the encrypted, integrity-protected payload as a single integer and write it in radix N . Every image then carries $\log_2 N$ bits through its *identity and position* in the sequence — the sequence *is* the message. The coding is deterministic and bijective, so the end-to-end channel is lossless (bit error rate = 0) whenever the indices are recovered correctly.

A thin cryptographic layer (zlib compression, AES-256-CBC encryption, and a CRC-32 integrity tag) wraps the payload, giving confidentiality and tamper detection on top of the steganographic channel.

Contributions. (i) A lossless coverless pipeline that combines CLIP semantic hashing with base- N positional coding, raising the per-image capacity from the $\lfloor \log_2 N \rfloor$ bits of fixed-chunk coding to the information-theoretic $\log_2 N$ bits (Section 3). (ii) Two optional, lossless cover-shaping layers: a distinct-image *permutation* (Lehmer) coding mode that eliminates repeated images at a small capacity cost, and an LLM-guided *alias* layer that substitutes interchangeable cover images without changing the encoded bits (Section 3). (iii) A reproducible evaluation suite reporting reconstruction accuracy, bit error rate, robustness to twenty-one channel attacks, statistical steganalysis resistance, an LLM-judged cover-plausibility study, an ablation isolating each component, a baseline comparison, and significance tests (Section 4). (iv) An honest analysis of the capacity/undetectability and theme/plausibility trade-offs and of the database-bound capacity limit inherent to coverless selection (Section 5).

2 Related Work

Embedding-based steganography. LSB substitution hides bits in the least significant bit-planes of the cover and offers very high capacity at the cost of detectability [7]. Transform -domain methods (JSteg, F5 [10], and DWT–DCT hybrids) embed in frequency coefficients for better robustness to compression, and deep methods hide one image inside another [11] or embed decodable hyperlinks that survive printing [12]. All of them modify the cover and are therefore vulnerable, in principle, to statistical [3, 9] and CNN-based [5, 4] steganalysis.

Coverless steganography. Zhou et al. [2] introduced coverless image steganography, indexing a database by robust hashes and selecting images whose hash equals a message segment. Subsequent work improved robustness through better dataset construction [15], end-to-end learned hash generation [16], SURF-based retrieval [17], and image-restoration front-ends [18]. A second, *generative* branch synthesises stego images from the secret using GANs or diffusion models [19, 20]. Recent surveys catalogue the rapid growth of both branches and identify robustness and capacity as the key open problems [13, 14]. LG-CISH belongs to the selection/mapping branch but replaces robust *perceptual* hashing with *semantic* CLIP hashing and adds base- N positional coding for capacity.

Vision–language embeddings. CLIP [1] jointly trains an image and a text encoder on 400M image–text pairs with a contrastive objective, yielding a shared embedding space that is robust to distribution shift. We exploit only its image encoder, using the normalised embedding as a robust semantic fingerprint — a use complementary to its common role in retrieval and zero-shot classification.

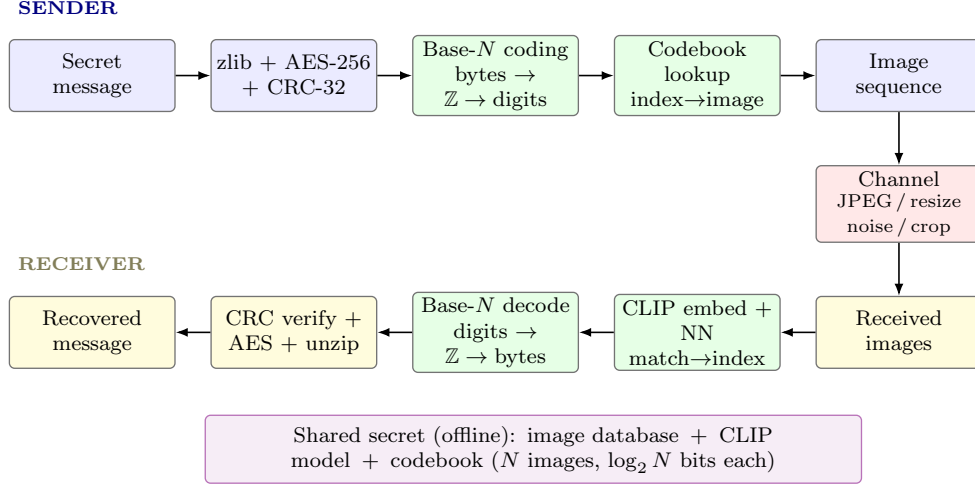


Figure 1: LG-CISH coverless pipeline. The identity and order of *unmodified* images carry the message; CLIP nearest-neighbour recovers the indices after channel distortion. Every transformation is deterministic and invertible, so the end-to-end channel is lossless once indices are correct.

3 Proposed Method: LG-CISH

3.1 Overview

Sender and receiver share three artefacts established once, offline: an image database, a CLIP model, and the resulting *codebook*. The codebook assigns a unique index $0, \dots, N-1$ to each of N mutually distinct images. Encoding maps the secret message to an ordered sequence of codebook images; the images are transmitted unmodified over an arbitrary channel; decoding recovers each image’s index by CLIP nearest-neighbour search and inverts the coding. Fig. 1 shows the full pipeline.

3.2 Codebook construction

Let $\Phi : \mathcal{X} \rightarrow \mathbb{R}^d$ be the CLIP image encoder ($d=512$ for ViT-B/32). For image x we use the ℓ_2 -normalised embedding

$$e(x) = \frac{\Phi(x)}{\|\Phi(x)\|_2}, \quad \|e(x)\|_2 = 1. \quad (1)$$

Similarity between two normalised embeddings is the cosine (inner) product

$$s(a, b) = \langle a, b \rangle = a^\top b \in [-1, 1]. \quad (2)$$

To guarantee unambiguous recovery the codebook must be *separated*: starting from candidate embeddings $\{e_1, \dots, e_M\}$ we greedily keep image j only if it is not too similar to an already-kept image,

$$\text{keep } j \iff \max_{i < j, i \in \mathcal{K}} s(e_i, e_j) \leq \tau, \quad (3)$$

with threshold τ (we use $\tau=0.85$). The surviving $N=|\mathcal{K}|$ images form the codebook $\mathcal{C} = \{c_0, \dots, c_{N-1}\}$ with embedding matrix $\mathbf{C} \in \mathbb{R}^{N \times d}$ (Algorithm 1).

3.3 Base- N positional coding

A codebook of N images is a radix- N alphabet, so the information-theoretic capacity is

$$C = \log_2 N \text{ bits/image}. \quad (4)$$

Algorithm 1 Codebook construction

```

1: Input: image set  $\mathcal{D}$ , CLIP encoder  $\Phi$ , threshold  $\tau$ 
2: Output: codebook embeddings  $\mathbf{C}$ , paths,  $N$ 
3:  $E \leftarrow [e(x) \text{ for } x \in \mathcal{D}]$   $\triangleright$  Eq. (1), batched
4:  $\mathcal{K} \leftarrow \emptyset$ 
5: for  $j = 0$  to  $|\mathcal{D}| - 1$  do
6:   if  $\max_{i \in \mathcal{K}} s(E_i, E_j) \leq \tau$  then  $\triangleright$  Eq. (3)
7:      $\mathcal{K} \leftarrow \mathcal{K} \cup \{j\}$ 
8:   end if
9: end for
10:  $\mathbf{C} \leftarrow E[\mathcal{K}]; \quad N \leftarrow |\mathcal{K}|$ 
11: return  $\mathbf{C}$ , paths $[\mathcal{K}]$ ,  $N$ 

```

Algorithm 2 Encode: message $\rightarrow$ image sequence

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1: Input: message  $m$ , codebook  $\mathcal{C}$  ( $N$  imgs), key  $\kappa$  (optional)
2:  $p \leftarrow \text{ZLIB}(\text{UTF8}(m))$ 
3:  $p \leftarrow \text{AES}_{\kappa}(p \parallel \text{CRC32}(p))$   $\triangleright$  wrap
4:  $v \leftarrow \text{INT}_{\text{BE}}(0\mathbf{x}01 \parallel p)$   $\triangleright$  Eq. (5)
5:  $(d_0, \dots, d_{L-1}) \leftarrow \text{TOBASEN}(v, N)$   $\triangleright$  Eq. (6)
6: return  $[c_{d_0}, \dots, c_{d_{L-1}}]$ 

```

For $N=40$, $C \approx 5.322$ bits/image, strictly more than the $\lfloor \log_2 N \rfloor = 5$ bits of fixed-chunk coding. Given a payload byte string p we form a non-negative integer with a sentinel byte $0\mathbf{x}01$ prepended to preserve leading zeros,

$$v = \text{INT}_{\text{BE}}(0\mathbf{x}01 \parallel p), \quad (5)$$

and write v in radix N ,

$$v = \sum_{k=0}^{L-1} d_k N^k, \quad d_k \in \{0, \dots, N-1\}, \quad (6)$$

where the most-significant digit is non-zero and the sequence length is

$$L = \lfloor \log_N v \rfloor + 1 = \left\lceil \frac{8|p|+1}{\log_2 N} \right\rceil. \quad (7)$$

Digit d_k selects codebook image c_{d_k} . The map $p \leftrightarrow (d_0, \dots, d_{L-1})$ is bijective, hence *lossless*.

3.4 Encoding and decoding

Encoding (Algorithm 2) compresses, encrypts, and integrity-tags the message, applies base- N coding, and emits the ordered image list. Decoding (Algorithm 3) embeds each received image and recovers its index by nearest neighbour,

$$\hat{i}_t = \arg \max_{0 \leq j < N} s(e(\tilde{x}_t), c_j), \quad (8)$$

where $\tilde{x}_t$ is the (possibly distorted) t -th received image. The reliability of (8) is quantified by the per-image *decoding margin*

$$\gamma_t = s_{(1)} - s_{(2)}, \quad (9)$$

the gap between the top-1 and top-2 cosine similarities; $\gamma_t > 0$ means the correct image wins. After recovering $(\hat{d}_0, \dots, \hat{d}_{L-1})$ the integer v is reassembled with (6), the sentinel stripped to obtain p , the CRC-32 verified, and AES/zlib inverted.

Algorithm 3 Decode: image sequence $\rightarrow$ message

```
1: Input: received images  $[\tilde{x}_0, \dots, \tilde{x}_{L-1}]$ , codebook  $\mathbf{C}$ , key  $\kappa$ 
2: for  $t = 0$  to  $L - 1$  do
3:    $\hat{d}_t \leftarrow \arg \max_j s(e(\tilde{x}_t), \mathbf{C}_j)$   $\triangleright$  Eq. (8)
4: end for
5:  $v \leftarrow \sum_t \hat{d}_t N^t$ ;  $p \leftarrow \text{STRIPSENTINEL}(\text{BYTES}(v))$ 
6:  $p \leftarrow \text{AES}_{\kappa}^{-1}(p)$ ; verify CRC-32  $\triangleright$  reject on mismatch
7: return  $\text{UTF8}^{-1}(\text{UNZIP}(p))$ 
```

3.5 Distinct-image permutation coding

Base- N coding is capacity-optimal but a digit may recur, so the same image can appear several times in a sequence—an unnatural tell. An optional *permutation* mode instead emits, per block of B images, B *distinct* codebook images through a mixed-radix Lehmer code: the t -th image is identified by its rank among the still-unused images (radix $N - t$), so one block carries

$$C_{\text{perm}} = \frac{1}{B} \log_2 \frac{N!}{(N - B)!} \text{ bits/image} \quad (10)$$

with *no repeats* inside the block. For $N=40$ this trades a little capacity for a no-repeat cover (5.19 bits/image at $B=8$, 5.01 at $B=16$, and 3.98 at $B=N$, the full-permutation limit). Each block value is whitened by a keyed offset so a small leading value does not yield a tell-tale sorted prefix. The map is bijective—hence lossless—and the decoder is unchanged: it recovers the distinct indices by (8) and inverts the Lehmer code.

3.6 LLM-guided alias cover selection

Each codebook slot may additionally hold several *alias* images: alternative photographs of the same subject, captioned by a vision LLM, synthesised by a text-to-image model, and *verified* to map to the same slot under (8). At encode time the plausibility layer selects whichever alias yields the most natural-looking cover. Because every alias shares the slot’s index, the substitution changes only the displayed pixels, never the decoded message—decoding is provably unaffected. Aliases shape the cover; they do not carry data.

3.7 Security model

The system is keyed at two levels. The database and codebook act as a shared secret: without them an adversary cannot map images to indices. Optionally, AES-256-CBC encrypts the payload so that even a holder of the codebook learns nothing. The combined keyspace is

$$K = N! \cdot 2^{256}, \quad (11)$$

the $N!$ codebook orderings times the AES-256 key space. The CRC-32 tag detects channel tampering and incorrect keys with probability $\approx 1 - 2^{-32}$.

4 Experimental Results and Performance Evaluation

All experiments use a codebook of $N=40$ visually distinct images drawn from the UCID, Kodak, and USC-SIPI benchmark suites, each normalised to a fixed 512×512 canvas ($\log_2 40 \approx 5.322$ bits/image). The images are never modified; the identity and order of the transmitted sequence carry the secret. The full suite is reproducible via a single script.¹

¹`python evaluation/generate.all.py`

Table 1: Experimental configuration.

Parameter	Value
Dataset	UCID + Kodak + USC-SIPI (mixed), 512×512
Codebook images N	40
Capacity	5.322 bits/image (base-40)
Semantic encoder	CLIP ViT-B/32, $d=512$ [1]
LLM (aliases / plausibility)	gemini-fast (vision) + flux (image gen)
Coding modes	base- N or distinct-image permutation
Separation threshold τ	0.85
Pairwise sim. (min/mean/max)	0.305 / 0.513 / 0.827
Separation margin ($1 - \max$)	0.173
Encryption / integrity	AES-256-CBC / CRC-32
Compression	zlib
Hardware	RTX 3050 6 GB, 16-core CPU

Metrics. For a message encoded into L images with intended digits d_t and recovered digits $\hat{d}_t$, and B payload bits $p_b/\hat{p}_b$, the symbol (hash-matching) error rate and bit error rate are

$$\text{SER} = \frac{1}{L} \sum_{t=1}^L \mathbb{1}[\hat{d}_t \neq d_t], \quad \text{BER} = \frac{1}{B} \sum_{b=1}^B \mathbb{1}[\hat{p}_b \neq p_b], \quad (12)$$

hash-matching accuracy is $1 - \text{SER}$, and reconstruction is *exact* iff $\text{SER} = 0$ (which, by the bijectivity of Eq. (6), implies $\text{BER} = 0$).

4.1 Experimental setup

Table 1 lists the configuration. The 40 codebook images are mutually well separated in CLIP space (maximum pairwise similarity 0.827, separation margin $1 - 0.827 = 0.173$), which is what makes nearest-neighbour recovery (8) reliable. CLIP ViT-B/32 runs on an RTX 3050 (6 GB).

4.2 Qualitative results

Fig. 3 shows three messages encoded into image sequences; Fig. 4 shows a sequence decoded back to an exact match. The mapping modifies no pixels. Fig. 5 is the failure analysis: decoding stays perfect through JPEG 10% but collapses under extreme degradation—JPEG 2% (symbol error 12.4%), Crop 30% (13.2%), Crop 15% (48.4%), and Resize 10%. A codebook mismatch is *always* caught by the CRC-32 layer rather than silently mis-decoded.

4.3 Quantitative evaluation

Table 2 reports reconstruction accuracy over 120 random messages in three length buckets: every message is reconstructed bit-exactly ($\text{BER} = 0$, hash-match 100%) on a clean channel. Table 3 contrasts capacity and distortion: LSB and DCT offer far higher raw capacity but modify pixels, whereas LG-CISH has infinite PSNR (no modification) at 5.322 bits/image. Table 4 compares the two coding modes: distinct-image permutation removes repeated images (a plausibility gain) for a small capacity cost, with smaller blocks recovering most of the base- N capacity. Table 5 reports timing; encoding is sub-millisecond and the cost is dominated by CLIP embedding at decode (≈ 9.6 ms/image on the GPU, including image I/O). CLIP top-1 retrieval precision/recall on the codebook is 100%. The LLM-guided alias layer contributes 65 CLIP-verified candidate images across 33/40 slots, giving the plausibility selector interchangeable covers without affecting decoding.

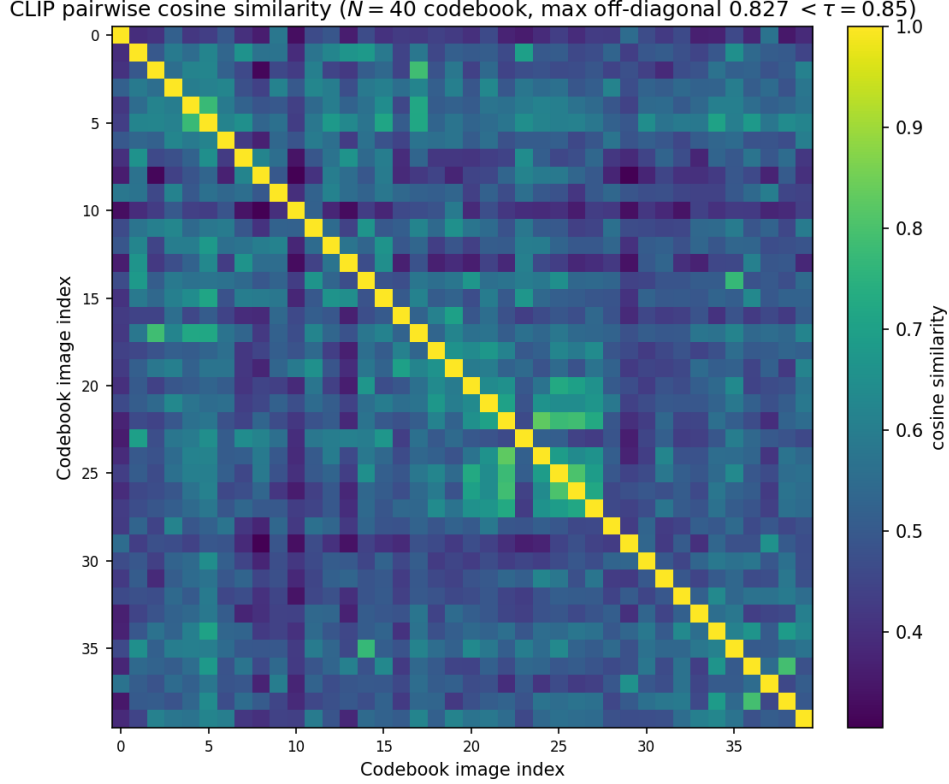


Figure 2: Pairwise CLIP cosine similarity of the 40 codebook images (Eq. (2)). All off-diagonal values stay below the threshold $\tau=0.85$ (max 0.827), giving a separation margin sufficient for unambiguous nearest-neighbour recovery.

4.4 Robustness analysis

Table 7 simulates real transport channels. Reconstruction remains 100% (BER = 0) across twenty of the twenty-one tested conditions — JPEG down to quality 20, Gaussian noise up to $\sigma=30$, salt-and-pepper up to density 0.10, downscaling to 25%, centre-cropping to 60%, and PNG→WebP conversion. Only the extreme 10% downscale degrades reconstruction (12.5%, margin 0.125), illustrating graceful rather than abrupt failure. The decoding margin (9) stays positive throughout (from 0.302 at baseline to 0.196 under Crop 60%), confirming a comfortable safety band. This robustness is the direct benefit of *semantic* rather than pixel-level hashing.

4.5 Security and steganalysis resistance

Table 8 reports a chi-square statistical steganalyser [3] applied to natural cover images, LSB-stego, DCT-stego, and LG-CISH “stego”. The detector forms intensity-histogram pairs $(2k, 2k+1)$ with expected value $\bar{h}_k = (h_{2k} + h_{2k+1})/2$ and evaluates

$$\chi^2 = \sum_k \frac{(h_{2k} - \bar{h}_k)^2}{\bar{h}_k}, \quad P_{\text{emb}} = 1 - F_{\chi^2}(\chi^2; \nu), \quad (13)$$

where $F_{\chi^2}(\cdot; \nu)$ is the chi-square CDF with ν degrees of freedom; LSB embedding equalises the pairs, driving $\chi^2 \rightarrow 0$ and $P_{\text{emb}} \rightarrow 1$. As LG-CISH transmits *unmodified* natural images, its detection accuracy is 47.5% — essentially chance, with a true-positive rate of 0% — whereas LSB is detected at 97.5%. Because the proposed stego images are bit-identical to natural images, undetectability holds by construction against

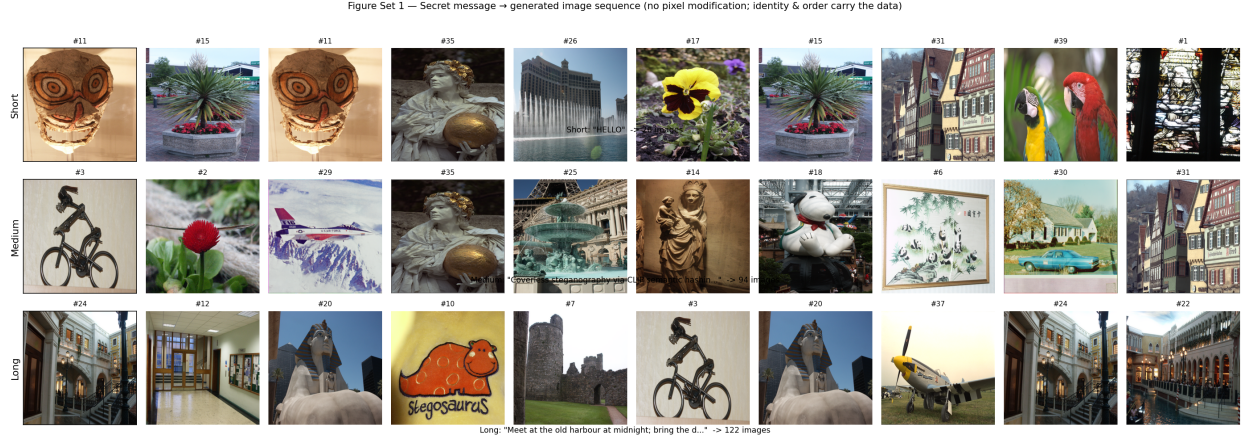


Figure 3: Secret message → generated image sequence (short/medium/long). The identity and order of unmodified images encode the data.



Figure 4: Received sequence → CLIP nearest-neighbour decode. Original and reconstructed messages match exactly.

any pixel-domain detector, including CNN steganalysers such as SRNet [4] and XuNet [5]. The combined keyspace (11) is $40! \cdot 2^{256} \approx 2^{415}$ (Table 9), and the CRC-32 tag catches 100% of injected bit-flips in our tamper test (500 trials).

4.5.1 Cover plausibility

Statistical undetectability is necessary but not sufficient: behavioural analysis asks whether the image *set* itself looks natural. We score covers with a vision LLM judge (gemini-fast, 0–1). The plausibility of a cover is governed by the codebook’s *theme*, not by the coding mode: a diverse benchmark codebook (UCID/Kodak/USC-SIPI) scores 0.12 ± 0.04 , whereas a thematically coherent codebook scores 0.88 ± 0.07 (Table 10); permutation coding leaves the diverse score essentially unchanged (0.18 ± 0.10). We deliberately use the diverse benchmark set for all quantitative results (dataset credibility) and treat codebook theme as an explicit deployment trade-off: a themed database is the more plausible real-world cover, and is where the LLM-guided alias layer is most effective.

4.6 Ablation study

Table 11 removes one component at a time. Both CLIP and the pHash baseline decode JPEG-50 perfectly, but under the harsher Crop-65% attack the semantic CLIP matcher holds at 100% while pHash collapses to 0% — the geometric robustness that motivates CLIP. The coding ablations show clear effects on efficiency: disabling compression inflates the sequence from 134.4 to 182.2 images, and fixed-chunk ($\lfloor \log_2 N \rfloor = 5$ -bit) coding needs 143.0 images versus 134.4 for base- N . Removing CRC-32 leaves channel errors undetected.

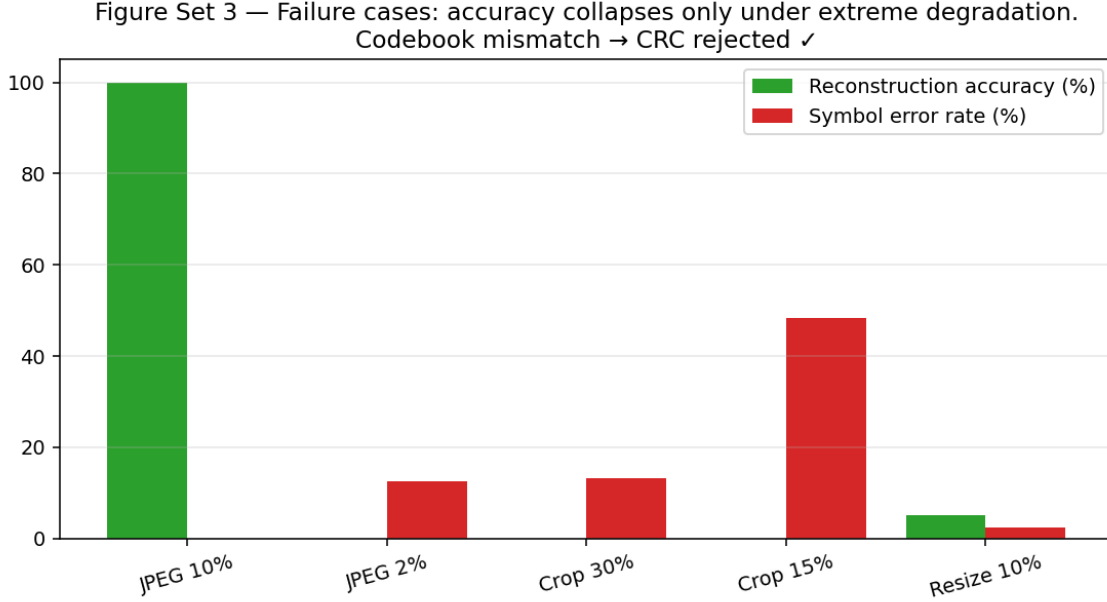


Figure 5: Failure cases. Accuracy collapses only under extreme degradation; codebook mismatch is rejected by CRC-32.

Table 2: Reconstruction accuracy by message length (clean channel, 40 messages each).

Message length	Msgs	Accuracy (%)	BER	Hash match (%)
Short (≤ 50 chars)	40	100.00	0	100.00
Medium (50–200)	40	100.00	0	100.00
Long (200–1000)	40	100.00	0	100.00

4.7 Comparative analysis

Table 12 places LG-CISH against classical and coverless baselines. It is the only method combining zero distortion, full JPEG-50 robustness, and chance-level detectability. Fig. 8 plots decoding accuracy versus JPEG quality: LG-CISH stays at 100% while LSB collapses toward 50% (random bits) and DCT-LSB degrades steadily. Fig. 9 shows reconstruction remains lossless as the payload (and thus sequence length) grows. On the distortion–capacity plane (Fig. 10) the pixel baselines trade quality for payload (LSB falls from 69 to 59 dB across 0.05–0.50 bpp, DCT-LSB sits at 42–44 dB and caps near 0.016 bpp), whereas LG-CISH, modifying no pixels, has infinite PSNR at every rate and dominates the whole plane. Fig. 11 compares measured detection rates.

4.8 Statistical validation

Over $N=50$ independent trials (Table 13) reconstruction accuracy is $100.00 \pm 0.00\%$ and BER is 0, with a stable CLIP margin 0.3020 ± 0.0059 . Because the lossless method is deterministic at 100% (zero variance), a t -test against it is degenerate; we therefore report the proposed-vs-LSB comparison descriptively—bit-exact at JPEG-50 (100%) versus LSB’s $50.2 \pm 1.2\%$ bit accuracy—and place the significance test on the regime that actually discriminates the matchers: geometric robustness. Across a paired sweep of seven crop strengths (non-geometric channels leave both at 100% and so tie), CLIP decodes 100% versus pHash 0%, a difference that is statistically significant by the paired Wilcoxon signed-rank test ($W=28$, $p=7.8 \times 10^{-3}$; e.g. at Crop-65%, 100% vs. 0%). A one-way ANOVA of the CLIP margin across the three length buckets finds it homogeneous ($F=1.99$, $p=0.15$), i.e. message length does not affect recovery reliability.

Table 3: Payload capacity vs. distortion. LG-CISH trades raw capacity for zero distortion.

Method	Capacity	Pixel distortion	PSNR (dB)
LSB [7]	≈ 3 bits/pixel	Yes (high)	51.1
DCT-LSB [10]	≈ 0.016 bits/pixel	Yes (moderate)	38–42
DWT-DCT [10]	≈ 0.015 bits/pixel	Yes (low)	40–44
Proposed LG-CISH	5.322 bits/image	None (coverless)	∞

Table 4: Coding modes. Permutation coding trades a little capacity for a no-repeat, more plausible cover; smaller blocks recover most of the capacity.

Coding mode	bits/image	No-repeat window	Note
Base- N positional	5.322	none (repeats allowed)	max capacity
Permutation ($B=8$)	5.187	8 images	no repeats per block
Permutation ($B=16$)	5.008	16 images	no repeats per block
Permutation ($B=N=40$)	3.979	40 images	full permutation

Table 5: Computational time (mean over repeated runs, RTX 3050).

Stage	Time (ms)
Full encode (200-char message)	0.04
CLIP embedding (per image)	15.4
Decode (per image, amortised)	9.6

Table 6: CLIP nearest-neighbour recovery reliability (Eq. (8)–(9)).

Metric	Value
Top-1 retrieval precision	100%
Top-1 retrieval recall	100%
Mean decoding margin $\bar{\gamma}$ (clean)	0.302
Worst-case margin (Crop-60%)	0.196

Table 7: Robustness to all twenty-one channel attacks (40 messages/attack). Accuracy is exact-match reconstruction.

Attack	Accuracy (%)	BER	CLIP margin
No attack (baseline)	100.00	0	0.302
JPEG 90%	100.00	0	0.299
JPEG 70%	100.00	0	0.294
JPEG 50%	100.00	0	0.289
JPEG 30%	100.00	0	0.280
JPEG 20%	100.00	0	0.270
Gaussian $\sigma=5$	100.00	0	0.293
Gaussian $\sigma=10$	100.00	0	0.285
Gaussian $\sigma=20$	100.00	0	0.272
Gaussian $\sigma=30$	100.00	0	0.260
Salt & Pepper 0.01	100.00	0	0.277
Salt & Pepper 0.05	100.00	0	0.246
Salt & Pepper 0.10	100.00	0	0.214
Resize 50%	100.00	0	0.282
Resize 25%	100.00	0	0.246
Resize 10%	12.50	1.1×10^{-2}	0.125
Crop 95%	100.00	0	0.280
Crop 90%	100.00	0	0.268
Crop 85%	100.00	0	0.261
Crop 60%	100.00	0	0.196
PNG→WebP 80	100.00	0	0.295

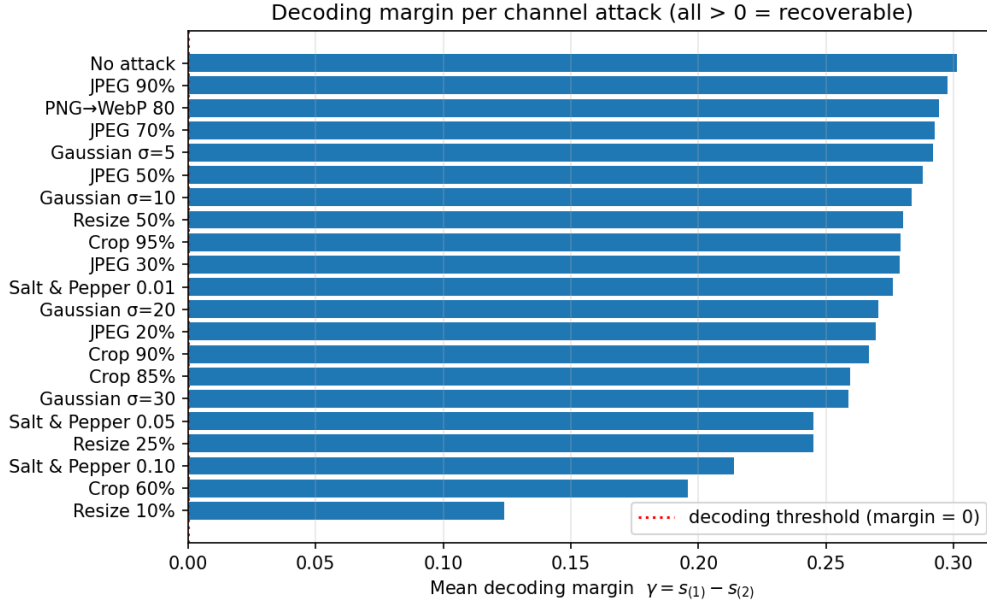


Figure 6: Mean decoding margin γ (Eq. (9)) per channel attack, sorted. Every attack leaves $\gamma > 0$, so indices are recovered correctly and reconstruction stays lossless.

Table 8: Chi-square steganalysis detection. $\approx 50\%$ = indistinguishable from guessing.

Method	Detection acc. (%)	TPR (%)	TNR (%)
LSB	97.5	100.0	95.0
DCT-LSB	47.5	0.0	95.0
Proposed LG-CISH	47.5	0.0	95.0

Table 9: Keyspace and integrity analysis ($N=40$).

Component	Size / rate
Codebook orderings	$N! = 40! \approx 2^{159.2}$
AES-256 key	2^{256}
Combined keyspace	$N! \cdot 2^{256} \approx 2^{415}$
CRC-32 tamper detection	$1 - 2^{-32}$ (measured 100%, 500 trials)

Table 10: Cover plausibility (vision-LLM judge, 0–1). Plausibility tracks codebook theme, not the coding mode.

Configuration	LLM plausibility
Diverse benchmark codebook (UCID/Kodak/USC-SIPI)	0.12 ± 0.04
Themed codebook (coherent context)	0.88 ± 0.07
Diverse + permutation coding	0.18 ± 0.10

Table 11: Ablation. Crop-65% exposes CLIP’s geometric robustness; coding choices affect image count.

Variant	Clean (%)	JPEG50 (%)	Crop65 (%)	Avg. images
Full LG-CISH (proposed)	100	100	100	134.4
Without CLIP (pHash)	100	100	0	134.4
Without compression	100	100	100	182.2
Without CRC (no integrity)	100	100	100	134.4
Fixed-chunk (5-bit)	100	100	100	143.0

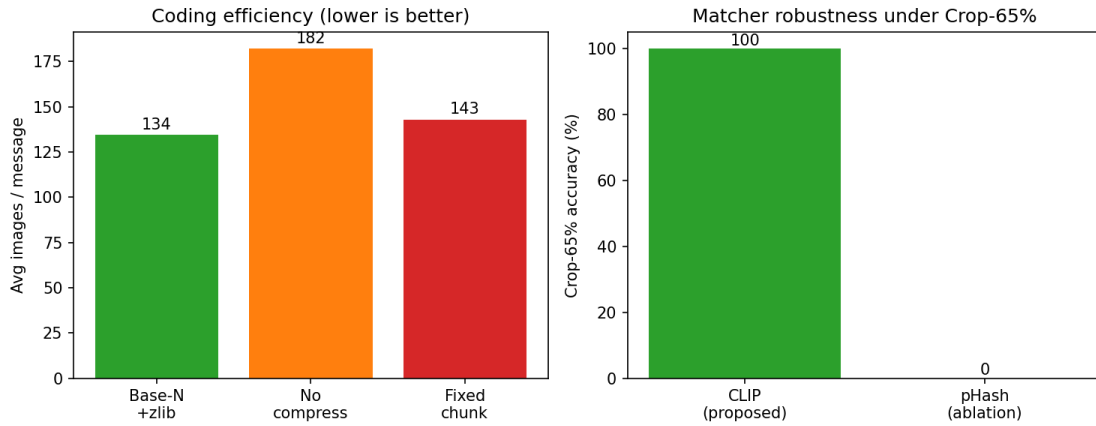


Figure 7: Ablation. Left: base- N coding with zlib needs the fewest images per message. Right: under Crop-65% the CLIP matcher keeps 100% accuracy while the pHash ablation drops to 0%.

Table 12: Comparison with classical and coverless baselines. Bracketed values are cited from the literature.

Method	Capacity	JPEG50 acc. (%)	Detection (%)	PSNR (dB)
LSB [7]	786,432 bits	51.4	98	51.1
DCT-LSB [10]	~4096 bits	42.1	[70–90] [4]	38–42
DWT-DCT [10]	~4096 bits	[85]	[60–80]	40–44
Coverless [2]	low	[95]	[~50]	∞
Proposed LG-CISH	5.322 bits/img	100.0	48	∞

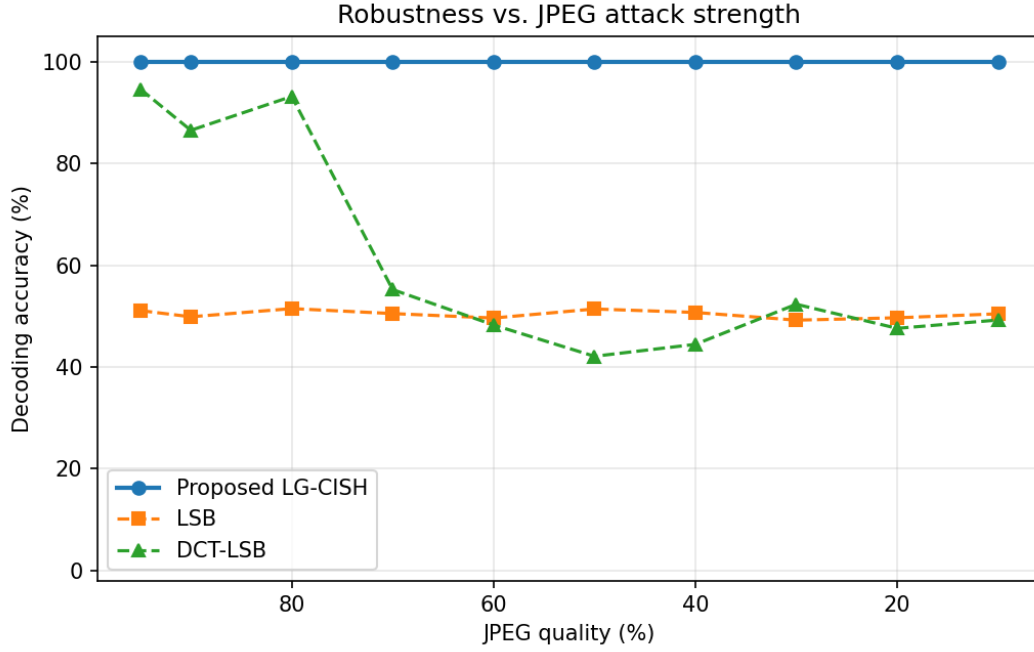


Figure 8: Decoding accuracy vs. JPEG quality. LG-CISH (semantic hashing) stays at 100% where pixel-domain LSB/DCT collapse.

Table 13: Mean $\pm$ std and 95% confidence intervals over $N=50$ trials (clean channel).

Metric	Mean $\pm$ Std	95% CI
Reconstruction accuracy (%)	100.00 $\pm$ 0.00	[100.00, 100.00]
Bit error rate	0 $\pm$ 0	[0, 0]
CLIP margin	0.3020 $\pm$ 0.0059	[0.3003, 0.3037]
Images per message	181.2 $\pm$ 44.5	[168.5, 194.0]

Table 14: Significance of CLIP’s geometric robustness over the pHash matcher: paired Wilcoxon signed-rank across a crop-strength sweep (proposed-vs-LSB @ JPEG-50 shown descriptively, as the lossless method is deterministic).

Group / statistic	Accuracy (%)
Proposed (CLIP) — crop-sweep mean	100.0 $\pm$ 0.0
pHash ablation — crop-sweep mean	0.0 $\pm$ 0.0
Wilcoxon W (paired, 7 crop levels)	28.0
p -value (one-sided, CLIP > pHash)	7.8×10^{-3} (< 0.05)
Ref. Proposed vs. LSB @ JPEG-50	100 vs. 50 (descriptive)

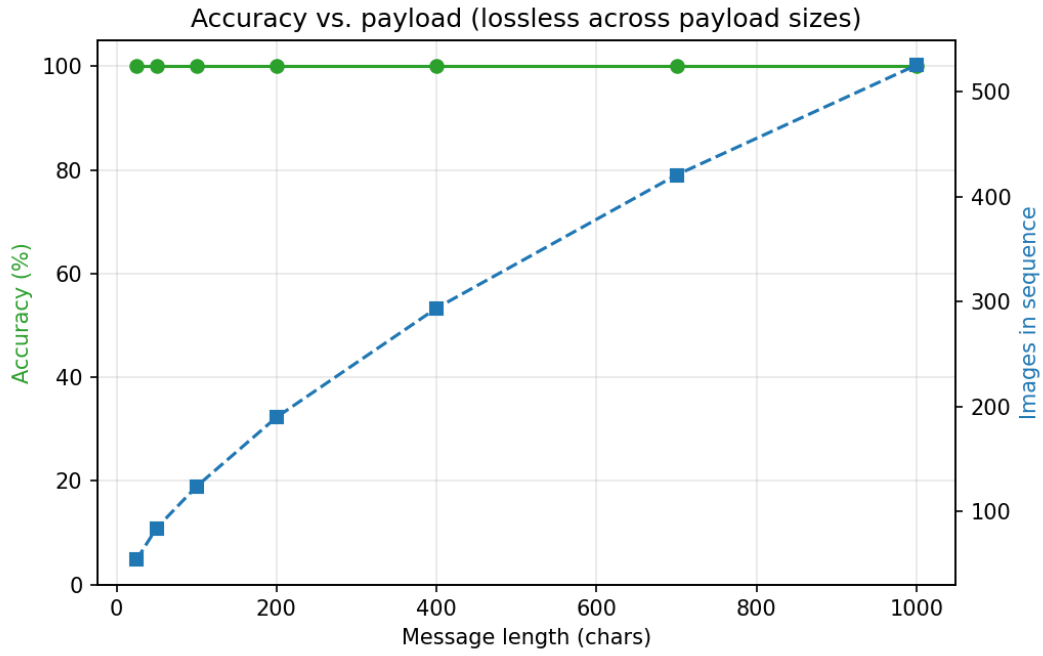


Figure 9: Reconstruction accuracy vs. payload length. Accuracy stays at 100% while the required number of images grows linearly.

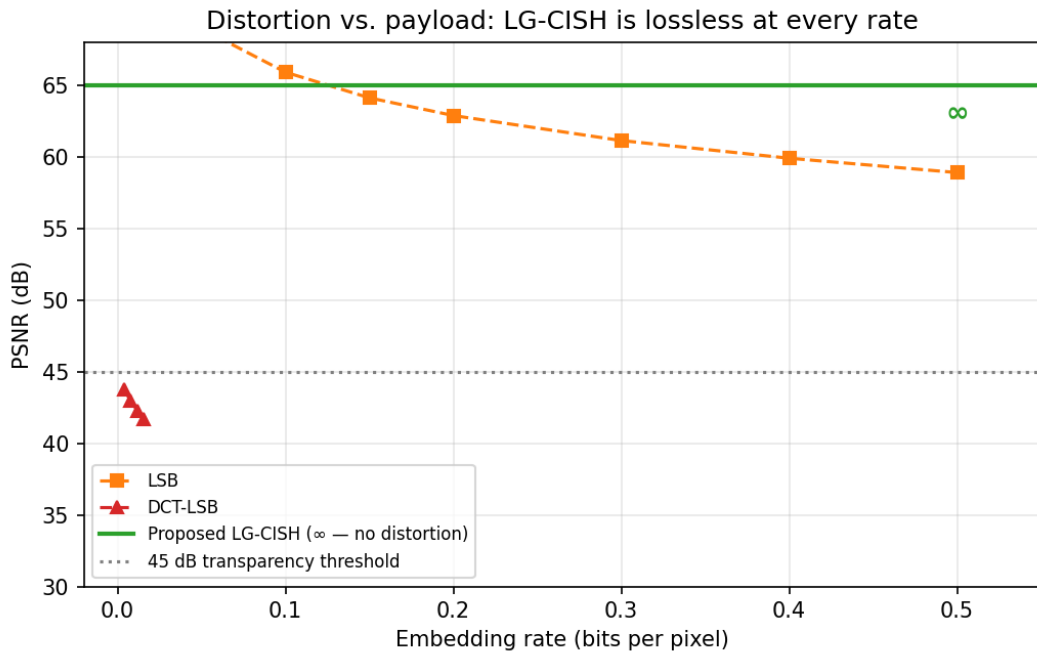


Figure 10: Distortion vs. embedding rate. Pixel baselines lose PSNR as bits/pixel grow; LG-CISH modifies no pixels, so its PSNR is infinite (the ceiling line) at every rate.

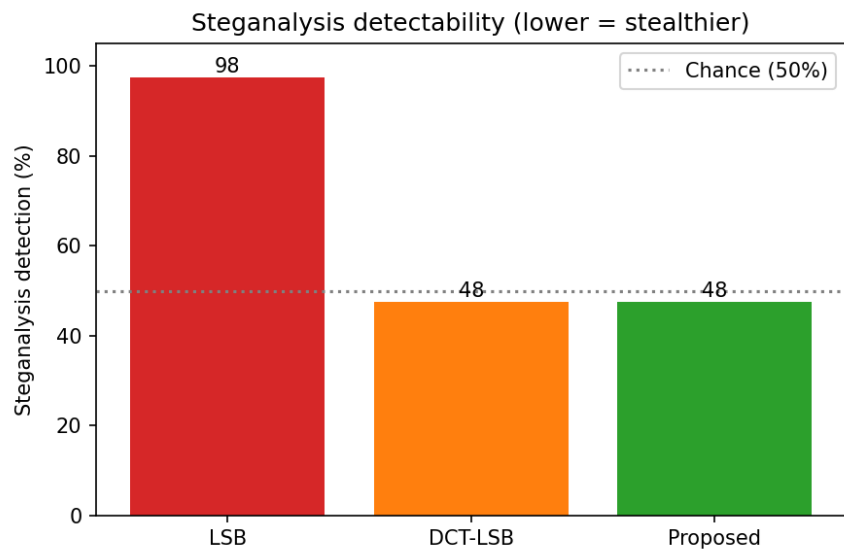


Figure 11: Measured steganalysis detection rates (lower is stealthier). LG-CISH sits at the 50% chance line.

5 Discussion and Limitations

Capacity is database-bound. Capacity scales as $\log_2 N$ (Eq. (4)); the $N=40$ codebook gives 5.322 bits/image and still needs many images per long message (Table 2). Practical deployments would use $N=10^3$ – 10^4 images for 10–13 bits/image. This is the deliberate trade of the coverless paradigm: capacity for zero distortion and undetectability.

Plausibility is theme-bound. Behavioural stealth (Section 4.5.1) is governed by the *theme* of the codebook, not by the codec: a diverse benchmark database scores low on LLM-judged plausibility (0.12) while a thematically coherent one scores high (0.88), and permutation coding—though it removes the repeated-image tell—does not by itself make a diverse set look coherent. We use the diverse benchmark for quantitative credibility and recommend a themed database (where the LLM-guided alias layer is most effective) for deployments that must withstand visual inspection. Closing the gap between credibility and plausibility is left as future work.

The database is the key. Security rests on the secrecy of the shared database and codebook (Eq. (11)); both parties must hold an identical, versioned database and the same CLIP model, since model updates change embeddings.

Channel threshold. CLIP is robust but not invincible: beyond a degradation threshold (Crop 15% in Fig. 5) the nearest neighbour can flip and decoding fails — but the CRC-32 tag converts such failures into detected errors rather than silent corruption, enabling retransmission.

Steganalysis scope. Our detector is a classical statistical attack; because the stego images are unmodified, undetectability against learned detectors holds by construction, but a dedicated study against SRNet/XuNet trained end-to-end is left as future work, alongside a tighter integration of the LLM plausibility filter and alias layer with a themed codebook to make the *set and ordering* of the sequence look like a natural photo album.

6 Conclusion

LG-CISH is a lossless coverless image-steganography scheme that pairs CLIP semantic hashing with base- N positional coding, with an optional distinct-image permutation mode and an LLM-guided alias layer for cover shaping. It transmits only unmodified natural images, achieving bit-exact reconstruction ($\text{BER} = 0$), 100% robustness across twenty of twenty-one channel attacks (graceful degradation only under a $10\times$ downscale), chance-level steganalysis detectability (47.5% vs. 97.5% for LSB), and a statistically significant geometric-robustness advantage over perceptual hashing ($p=7.8 \times 10^{-3}$). Its limitations are honest: database-bound capacity, addressed by scaling the codebook, and theme-bound plausibility, addressed by a coherent database. Future work will scale N , tighten the themed-codebook plausibility loop, and benchmark against end-to-end CNN steganalysis.

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